

Course Syllabus

CHE 253K Applied Statistics (Unique Number 16525)

Spring 2026

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Schedule

Lectures: MW 9-10 AM, PMA 6.104; Recitation F 9-10AM, PMA 6.104; note that in some weeks a lecture and recitation may be switched. This will be announced well in advance.

Code of Conduct

1. **Attendance** is highly recommended.

This class aims to provide an open discussion environment, in which every student engages and contributes to the learning process. Consequently:

2. **Cell phones** should be placed in “silent” mode and put away out of sight (preferably in your bag). You do not need them during the class period and I find them detrimental to the instruction process.

3. **Laptops** and **tablets** may be used for note taking purposes or when needed as communicated by the instructor. Please do not multi-task during class.

Overview of the Course and Learning Objectives

ChE 253K Applied Statistics is a required course that introduces the tools of statistical analysis, with the purpose of representing, understanding and interpreting data collected from various sources, such as laboratory or plant measurements, surveys, etc. These analyses ultimately serve to make decisions that can affect product development, process operations, capital investment, etc.

The knowledge, abilities and skills that students are expected to gain from this course are:

- Describe and represent graphically a set of data
- Define and apply the basic rules of probability and probability distributions
- Formulate and test statistical hypotheses

- Generate linear and nonlinear regression models from data
- Design an experimental plan
- Deploy these skills in software such as Excel and JMP

These learning objectives fall within the broader scope of ABET Objectives 1 and 6:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions

Prerequisites: ChE 210 (Introduction to Computing), and M 427J (Differential Equations). **Co-requisite:** ChE 317 (Introduction to Chemical Engineering Analysis).

Course materials:

Textbook: *Probability and Statistics for Engineering and the Sciences*, 9th ed. Jay L. Devore, Cengage Learning, Boston, MA. (ISBN: 9781305251809)

NOTE: The textbook this class is available through the Longhorn Textbook Access (LTA) program, an initiative between UT Austin, The University Co-op and textbook publishers to significantly reduce the cost of digital course materials for students. You are automatically opted into the program but can easily opt-out (and back in) via Canvas through the 12th class day. To opt out, simply press the “My Textbooks” link in Canvas and then press the “Opt Out” button on the material. You will be billed for your LTA material after the 5th class day through your “What I Owe” page but still have through the 12th class day to opt out. You must pay for your LTA materials through the What I Owe page by the 20th class day. If you do not pay by the 20th class day, you will lose access to the materials after the 20th class day and your charge will be removed. More information about the LTA program is available at [Longhorn Textbook Access – University Co-op](#). Information on the VitalSource off-line reader can be found [here](#).

Other materials will be provided in print or electronic form as necessary.

Structure

The lectures will introduce key concepts (see Schedule, below), which will be supported with examples. Recitation sessions will be dedicated to solving problems based on these concepts.

Grading

The course grade will account for student performance in i) quizzes, ii) homework, iii) mid-term and final exams

- i) Homework will be assigned regularly and will be due within a week. There will be a 25% per calendar day grade penalty for turning the homework in late (unless prior approval from the instructor has been obtained). The homework will be graded following the solution key and rubric, and the final score will be reduced as described above.

You are encouraged to discuss the assignments with each other; however, each student must turn in their own solution.

- ii) Quizzes: will be administered via UT Instapoll during the lecture and recitation periods. The correct answers will be awarded one point to count towards the final grade, the wrong answer will be awarded 0.5 points, no answer will receive no credit. The top 10 quiz scores will count towards the final grade, as described below.
- iii) Exams: there will be two mid-semester exams during regular lecture hours, and one final exam (the date for the final is set by the Registrar).

The dates for the mid-semester exams are:
Mid-semester 1: Monday February 23, 9-10 AM
Mid-semester 2: Monday April 6, 9-10 AM

The final grade will be computed as follows:

Homework (averaged), total contribution	20%
Quizzes	10%
Mid-semester exams, 20% each	40%
Final exam	30%
Total	100

Regrade requests for any assignments: please submit regrade requests **in writing** to the instructor. Please write your request on a separate sheet of paper and attach it to the original graded assignment; do not make any changes or annotations to the original document. Regrade requests must be submitted within one calendar week (7 calendar days) of receiving your graded assignment. The day the graded assignment was received counts as day 1.

Extra credit

From time to time, I may offer opportunities to earn extra credit, either individually or for the entire class. Individual opportunities might include being the first to correctly answer a challenging question in class or completing an optional homework problem. Extra credit may also be awarded to the whole class for strong participation in surveys (such as informal course surveys or the official university course evaluation), if a participation threshold is met (for example, 80% of enrolled students responding).

Each extra-credit opportunity is worth up to 1 point and is applied to the final exam score before the final course grade is calculated. The final exam score is capped at 100 points, including extra credit. For example, if you earn a 99 on the final exam and receive 1 extra-credit point, your final exam score will be 100. But, if you earn 100 on the final exam, your final exam score will remain 100.

Letter grades will be assigned as follows:

1. The weighted average score (based on quizzes, homework and all exams, weighted as above) will be scaled by the highest average score in the class.

Example: assume the highest average score is 95. All scores will be divided by 0.95 and rounded up to two significant digits. This way, 95 becomes 100, 90 becomes 94.74, etc. I will communicate the highest score when it becomes available.

2. Letter grades will be allocated based on the scaled scores:

Scaled score		Letter
From	to	Grade
0	59.99	F
60	62.99	D-
63	66.99	D
67	69.99	D+
70	72.99	C-
73	75.99	C
76	79.99	C+
80	82.99	B-
83	85.99	B
86	89.99	B+
90	92.99	A-
93	100	A

Plus/minus grading will be used.

Accommodations

“The university is committed to creating an accessible and inclusive learning environment consistent with university policy and federal and state law. Please let me know if you experience any barriers to learning so I can work with you to ensure you have equal opportunity to participate fully in this course. If you are a student with a disability, or think you may have a disability, and need accommodations please contact Disability and Access (D&A). Please refer to D&A’s website for contact and more information: <http://disability.utexas.edu/>. If you are already registered with D&A , please deliver your Accommodation Letter to me as early as possible in the semester so we can discuss your approved accommodations and needs in this course.”

Absences

Students must provide a certified medical excuse letter or a letter on University letterhead (for school-related activities) in order to reschedule any assignment or change/extend a deadline.

Academic Integrity

Academic integrity violations will result in a 0 (zero) grade for the respective exam, quiz or assignment, and an “F” grade for the entire course.

Forms of scholastic dishonesty include (but are not limited to) plagiarism, use of unauthorized materials for exams, and changing solutions to returned assignments, quizzes and exams.

The emergence of generative AI tools (such as ChatGPT and Copilot) has sparked interest in all science and engineering fields. The use of these tools for brainstorming ideas, exploring possible responses to questions or problems, and creative engagement with the materials may be useful for you as you craft responses to class assignments. While there is no substitute for working directly with your instructors (including lecturer, TA, tutor), the potential for generative AI tools to provide automatic feedback, assistive technology and language assistance is clearly developing. Keep in mind that the answers provided by these tools may or may not be correct or factual, and that you are ultimately responsible for your assignment submissions. You should make it clear IF, WHERE and HOW you used generative AI in any material that you submit in response to class assignments. Please feel free to reach out to me well in advance of the due date of assignments for which you may be using generative AI tools and I will be happy to discuss what is acceptable. (adapted from <https://ctl.utexas.edu/chatgpt-and-generative-ai-tools-sample-syllabus-policy-statements>).

Note that ALL exams are CLOSED-BOOK, CLOSED NOTES. That means that you will NOT be able to bring any course materials with you for the exam, and will not have access to generative AI.

Course schedule

There will be two lectures each week (except exams). The schedule below is tentative and subject to change based on student progress:

Lecture No.	Topic
1	Intro, course objectives, syllabus review, basics
2-3	Descriptive statistics
4	Basic Rules of Probability
5	Solving probability problems with tree diagrams
6	Conditional probabilities and Bayes Theorem
7	Discrete probability distribution functions
8	Binomial distribution function and applications
9	Continuous probability distribution functions & The Normal Distr.
10	The t-distribution and its properties & the Central Limit Theorem
11	Combinations of Distributions
12-13	Confidence intervals

14-17	Hypothesis Testing
18-20	Statistical process control
21-24	Linear and non-linear regression
25-27	Design of Experiments/Response Surface Models
28	Course review